

SPECIFICATION — NODE 1 — PAGE 4

EXTERNAL LONGITUDINAL WEARABLE — NODE-SPECIFIC EXEMPLAR CLAIM-LIKE STATEMENTS (OPTIONAL, NON-LIMITING)

A. Overview

This page provides optional, non-limiting claim-like statements for Node 1 to support standalone protectability of the external longitudinal wearable. A provisional application does not require claims. These statements are illustrative only and do not limit the disclosure.

B. Exemplary Node 1 System Statements (Non-Limiting)

1. A wearable device comprising: a) a ring-shaped or C-shaped housing configured for prolonged wear during sleep and/or daily life; b) at least one sensing modality selected from pressure sensing, strain sensing, magnetic displacement sensing, optical sensing, temperature sensing, and inertial sensing; and c) a processor and/or a communications interface configured to provide time-series data to a computing system; wherein the computing system computes at least one quality index and outputs privacy-preserving metrics representing longitudinal physiologic patterns. 2. The wearable device of statement 1, wherein the magnetic displacement sensing comprises a hall-effect sensor and a magnet configured to measure relative displacement indicative of deformation or coupling change. 3. The wearable device of statement 1, wherein the optical sensing comprises photoplethysmography (PPG) configured to provide physiologic confirmation and/or pulse feature extraction. 4. The wearable device of statement 1, wherein the inertial sensing comprises a multi-axis IMU configured to support motion gating, orientation estimation, and artifact rejection. 5. The wearable device of statement 1, wherein the computing system defines valid windows by applying threshold criteria to at least two quality index components comprising contact integrity and motion stability. 6. The wearable device of statement 1, wherein privacy-preserving metrics comprise one or more of: rigidity index, tumescence pattern index, duration index, stability/variability index, thermal context tags, and confidence indicators. 7. The wearable device of statement 1, wherein the wearable is configured for duty-cycled sampling and event-triggered burst sampling based on detected transitions in at least one sensor modality. 8. The wearable device of statement 1, wherein the wearable includes a compliant inner interface configured to maintain stable coupling while reducing localized pressure points during prolonged wear. 9. The wearable device of statement 1, wherein the computing system performs anti-spoof confidence scoring by requiring coherence across at least two modalities comprising pressure/contact sensing, magnetic displacement sensing, optical sensing, thermal sensing, and inertial sensing.

C. Exemplary Node 1 Method Statements (Non-Limiting)

10. A method comprising: a) acquiring time-series data from an external wearable configured for prolonged wear; b) computing at least one quality index comprising contact integrity and motion gating; c) selecting valid windows based on the quality index; and d) generating privacy-preserving longitudinal outputs comprising indices and confidence indicators. 11. The method of statement 10, further comprising computing a rigidity index using pressure and/or strain sensing and magnetic displacement sensing during the valid windows. 12. The method of statement 10, further comprising using PPG-derived pulse features as physiologic confirmation for at least a subset of windows. 13. The method of statement 10, further comprising outputting a stability index describing repeatability across multiple valid windows collected across multiple days or nights. 14. The method of statement 10, wherein data windows failing a threshold are down-weighted or excluded and labeled as reduced confidence.

D. Optional Integration Statements (Non-Limiting)

15. The system or method of any prior statement, further comprising time-aligning derived metrics from the external wearable with derived metrics from at least one additional node and computing a cross-device alignment score. 16. The system or method of any prior statement, further comprising gating cross-user correlation outputs under consent controls and requiring enhanced quality thresholds prior to including windows in correlation computations.

E. Variations

Any statement herein may be modified to include additional sensors, fewer sensors, substituted equivalent modalities, different housing geometries, different duty cycles, and different processing distributions (on-node, on-device, cloud) while maintaining the functional intent of longitudinal capture, validity gating, and privacy-preserving outputs.